

Aarushi Nema

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Experience

Data Analyst, IT Data & Analytics, Deloitte Singapore Oct 2025 – Present

- Developed SQL, Python, Excel, and Microsoft Power Platform workflows across 10+ Audit & Assurance engagements to automate revenue recomputation, invoice/vendor reconciliation, journal entry testing, variance analysis, exception flagging, financial statement and XBRL reconciliation, and dashboard data preparation
- Built Power BI and Tableau dashboards connected to SQL, Power Query, and Excel sources across 5+ Audit & Assurance engagements, enabling stakeholders to drill down into revenue variances, exception flags, control implementation status, risk indicators, and testing lifecycle progress
- Developed AI-enabled automation and frontend solutions, including an n8n-based sector and client news monitoring workflow that retrieved RSS articles, used LLMs to identify top relevant news and generate WhatsApp-ready summaries, and AWS-hosted frontend interfaces for internal chatbot, balance sheet / P&L variance analysis, and automated financial statement preparation tools
- Implemented internal automation and GRC solutions using Power Apps and Power Automate, digitizing audit workflows, standardizing data collection, and supporting governance, risk, and compliance tracking

Data Platform Intern, Hyundai Motor Group Innovation Centre Singapore May 2024 – Aug 2024

- Engineered and optimized scalable data migration pipelines from Relational DBMS (PostgreSQL and Tiberio) to Hadoop Data Lake using Python, PySpark, Apache Airflow, and Bash scripting
- Developed RESTful APIs using Flask and PySpark to enable dynamic data retrieval and filtering from Hadoop Data Lake across multiple data formats

Software Development (Data Application) Intern, Infineon Technologies Singapore May 2023 – Dec 2023

- Collaborated with stakeholders and product engineers to address production data challenges, enhancing Tableau dashboards and developing Python ETL scripts to automate production yield record loading into a SQL database
- Spearheaded development of a centralized Robotic Process Automation (RPA) solution utilizing UiPath and Python, resulting in a 90% reduction in human intervention across four critical software tools
- Designed and implemented a Confluence-based chatbot using React, Flask, and SQL, significantly reducing time spent searching for and navigating through various software tools

Publications & Conferences

- **A. Nema**, X. Zhou, A. J. Jillella, S. Zhao, and S. Liu. “Context-Aware and Knowledge-Grounded Conversational Recommendation with Prompt Learning.” In **Pacific Rim International Conference on Artificial Intelligence (PRICAI 2025)**, Springer Nature. https://link.springer.com/chapter/10.1007/978-981-95-7078-2_26
- A. Balaji, **A. Nema**, N. Kamath, and J. Raju. “Encoding Values: Injecting Morality into Machines via Prompt-Conditioned Moral Frames.” In **NeurIPS 2025 Creative AI Track**. Published on OpenReview. <https://openreview.net/forum?id=gzX840IFM4>

Education

Nanyang Technological University, Singapore Aug 2021 – Jun 2025

Bachelor of Computing, Data Science and Artificial Intelligence

CGPA: 4.22/5.00 (Honors with Distinction)

Projects

Final Year Project: UniCRSxGraphRAG (Grade: A) GitHub: Project-UniCRSxGraphRAG

LLMs | Knowledge Graphs | Recommender Systems | Python | PyTorch | GraphRAG | OpenAI

- Architected and implemented an end-to-end conversational recommender system integrating pretrained large language models (DialoGPT) with GraphRAG for recommendation optimization and explainability

Machine Learning - Elo Merchant Category Recommendation Kaggle Competition (Placed in top 5) GitHub: Elo-competition

LightGBM | Feature Engineering | Ensemble Learning | PyTorch

- Engineered a top 3.2% winning solution for the Elo Merchant Category Recommendation task using advanced feature selection (SHAP, RFE, RFM clustering), outlier-aware modeling, and LightGBM ensembles

- Built a data analysis pipeline (cleaning, transformation, feature engineering) on global life expectancy datasets, applying statistical methods and correlation analysis to uncover socioeconomic drivers (GDP, healthcare, education) and temporal trends to create visualizations (heatmaps, time-series plots, choropleth maps)

Skills

Data Analytics & BI: SQL, Excel, Power BI, Tableau, Power Query, Data Visualization

Data Engineering & Platforms: PySpark, ETL Pipeline Development, PostgreSQL, MongoDB, Hadoop Data Lake, Apache Airflow, Jenkins, Data Migration, Data Synchronization

Programming & Application Development: Python, Java, Bash, JavaScript, ReactJS, React Native, NodeJS, Flask, HTML, CSS, REST APIs, Linux

Automation & AI Tools: Microsoft Power Platform (Power Apps, Power Automate), UiPath, n8n, LLMs (OpenAI, Anthropic, HuggingFace), Workflow Automation, AWS

Machine Learning & AI: PyTorch, Scikit-learn, TensorFlow, LightGBM, Feature Engineering, Recommender Systems, Knowledge Graphs, GraphRAG, Prompt Engineering

Tools & Collaboration: Git, Docker, Jira, Confluence, Postman, Figma, PowerPoint, Stakeholder Communication, Requirements Gathering

Co-Curricular Activities

- Vice-Chairperson**, Information Technology Committee, NTU Student Union

Aug 2022 – Aug 2024

 - Collaborated with external stakeholders and led a team of UI/UX designers to create Figma designs for new projects, including a locker booking portal and course timetable planner for students
- INSPIRE Women in Tech Programme Mentee**, Credit Suisse (Singapore)

Apr 2023 – Jul 2023

 - Completed a full-stack mobile app solution to increase awareness of unconscious bias under the mentorship of Credit Suisse’s technology team